

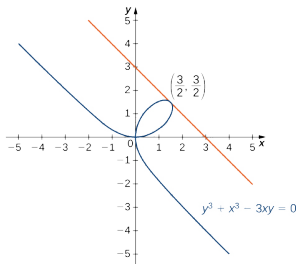
Week 05, Lecture 2

Implicit Differentiation; Exponential Functions

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Today, in Engineering Calculus...

- 1 Implicit Functions
- 2 Implicit Differentiation
- 3 Exponential functions
- 4 Properties of exponents
- 5 Exercises

Slides are on canvas, and
at [niallmadden.ie/2627-
MA140](http://niallmadden.ie/2627-MA140)

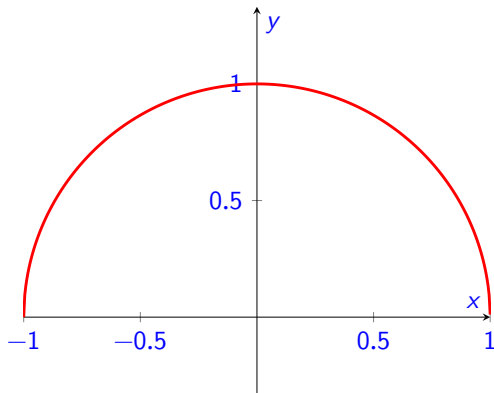


See also: Sections 3.8 (Implicit Differentiation) and 3.9 (Derivatives of Exponential and Logarithmic Functions) of **Calculus** by Strang & Herman: <https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation> and <https://openstax.org/books/calculus-volume-1/pages/3-9-derivatives-of-exponential-and-logarithmic-functions>

Implicit Functions

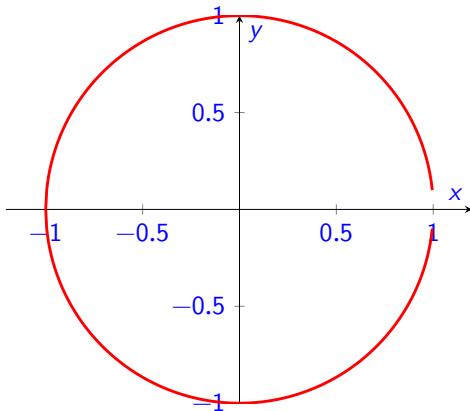
To date, most functions we have studied have been **explicitly** defined. Such functions can be written as $y = f(x)$: given a value of x we can substitute it into $f(x)$ to get the corresponding value of y .

Example: $y = \sqrt{1 - x^2}$.



Implicit Functions

However, sometimes we are given an equation involving x and y where these two terms are not “separated” entirely; e.g., $x^2 + y^2 = 1$. Here y is **implicitly** defined: for any pair (x, y) we can check if it is on the curve described by the equation.



Implicit Differentiation

Implicit equations define curves; we can use a technique called **implicit differentiation** to find tangents to these curves.

Method:

1. Differentiate both sides of the equation, with respect to x , keeping in mind that y is a function of x , using the Chain Rule where needed.
2. Solve for dy/dx .

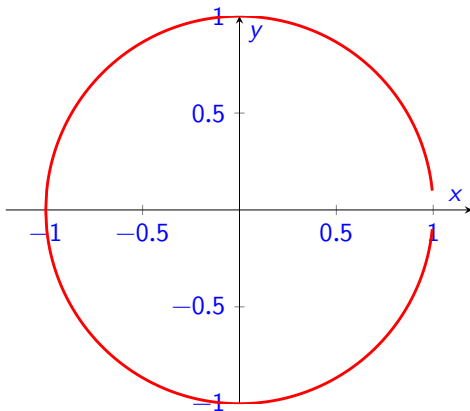
Implicit Differentiation

Example

If y is defined by $x^2 + y^2 = 1$, find $\frac{dy}{dx}$.

Implicit Differentiation

Now we know that if $x^2 + y^2 = 1$, then $\frac{dy}{dx} = -\frac{x}{y}$. We can check that this relates to the slope of the tangents to this curve at various places:



Implicit Differentiation

Find the tangent to the curve $x^2 + y^2 = 25$, at the point $(3, -4)$.

First check that the point $(x, y) = (3, -4)$ is on the curve:

$$(3)^2 + (-4)^2 = 9 + 16 = 25, \text{ as required.}$$

Now differentiate:

$$\begin{aligned}\frac{d}{dx}[x^2 + y^2] &= \frac{d}{dx}[25] &\implies \frac{d}{dx}[x^2] + \frac{d}{dx}[y^2] &= 0 \\ &&\implies 2x + 2y \frac{dy}{dx} &= 0 &\implies \frac{dy}{dx} &= -\frac{x}{y}.\end{aligned}$$

So the slope of the tangent to the curve at $x_1 = 3$ and $y_1 = -4$ is $M = -x/y = 3/4$.

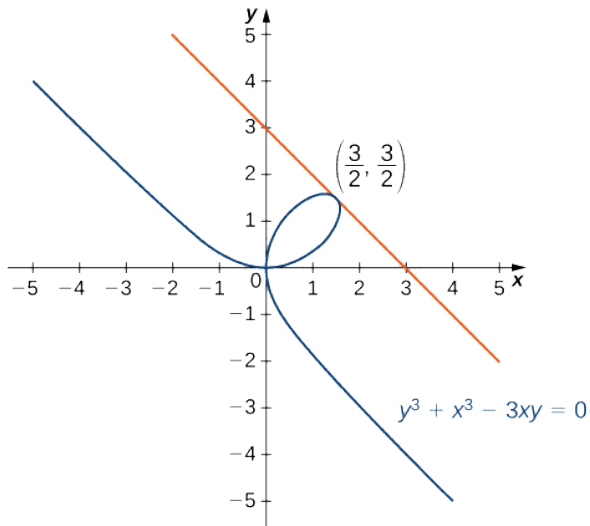
The equation of the tangent comes from $y - y_1 = M(x - x_i)$. For our values of $x_1 = 3$, $y_1 = -4$ and $M = 3/4$, we get

$$y = (-4) + (3/4)(x - 3). \text{ That is } y = \frac{3}{4}x - \frac{25}{4}.$$

Implicit Differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

Implicit Differentiation



Exponential functions

Earlier in this course we met functions such as $y = x^2$; this is a **power** function.

Now we consider **exponential functions**, such as $y = 2^x$.

Such functions occur in many applications. For example: if I invest €100 with an annual interest rate of 20%, then after x years, I will have $€100 \times (1.2)^x$. **Why?**

Exponential functions

Exponential functions grow quite fast: if my investment is indeed worth $f(x) = 100 \times (1.2)^x$ euros after x years, then...

- ▶ After 1 year, I have €120
- ▶ After 10 years, I have €619.17
- ▶ After 20 years, I have €3,833.80
- ▶ After 25 years, I have €9,539.60
- ▶ After 50 years, and 190 days, I'll be a millionaire!

Properties of exponents

Here I remind you of some properties of exponents that you should already know: for any positive numbers a and b , and for all x and y ,

$$1. \quad b^x b^y = b^{x+y}$$

$$2. \quad \frac{b^x}{b^y} = b^{x-y}$$

$$3. \quad (b^x)^y = b^{xy}$$

$$4. \quad (ab)^x = a^x b^x$$

$$5. \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

There is more information on this in [Section 1.5 of the textbook \(Exponential and Logarithmic Functions\)](#). ([← link to book](#))

Exercises

Exercise 5.2.1

Find the equation of the tangent to the curve defined by $x^2 - y^2 = 16$ at the point $(5, 3)$.

For more exercises on Implicit Differentiation, see <https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation>